

Ankur Desai

(248) 657 – 3805 | ardusa05@gmail.com | ardusa.github.io | linkedin.com/in/ardusa

EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding by automating provisioning (GitHub API + Slack SDK), granting repo and workspace access in 30s and cutting setup time 97% across 100+ members.
- Replaced paper sign-in with a QR-driven check-in backed by AWS S3 token caching, cutting processing time 80%.
- Reduced bug escape 65% via GitHub Actions CI/CD (90% coverage, 20+ tests), streamlining debugging, troubleshooting, and deployment.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Built Spring Boot backend with PostgreSQL/SQL and Neo4j on a cross-functional 6-engineer team, cutting p95 latency 48% (850ms → 440ms) and eliminating recursive SQL join overhead.
- Refactored Cypher with relationship-type indexing; traversals 3× faster than SQL self-joins, removing full-table scans.
- Integrated OAuth 2.0 (Google + GitHub) with httpOnly JWT cookies, refresh rotation, and user-scoped graph access.
- Implemented GitLab CI/CD for full lifecycle (build, test, deploy) with Docker and branch protection, enforcing 85% coverage across 35+ tests.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery* May 2025 – Nov. 2025

- Engineered multi-client voice assistant that resolves implicit speaker intent and dispatches real tasks across concurrent streams.
- Reduced API response time 73% by offloading RAG to async Redis/Celery queue.
- Built Electron + Next.js client with OAuth interception and IPC/webhook server for sub-50ms event propagation.
- Deployed FastAPI on AWS Lambda with lazy-loaded ML models, reducing cold-start 86% (6,643ms → 930ms).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

- Won Interactive Media Track: gesture-controlled wizard duel with full-body spells and abilities vs. AI opponent.
- Built MediaPipe Pose pipeline mapping 33 landmarks to game actions; 60 FPS, sub-20ms input lag.
- Built 2D physics engine (AABB collision) mapping 3D skeletal coordinates to game state; <5px drift at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, Kotlin, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash

Frameworks Spring Boot, FastAPI, Node.js, Angular, React, Next.js, Redis

Databases PostgreSQL, Neo4j, MongoDB

Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitLab CI/CD, CI/CD Pipelines